

Discussion 8: Relativistic energy and momentum – Solutions

Names:

Section:

This week we will explore some direct experimental justifications for the relativistic energy and momentum formulas derived in class: $p = \gamma mv$, $E = \gamma mc^2$, and $E_0 = mc^2$.

Part 1: Newtonian mechanics is incomplete!

In 1962, an experiment was performed to illustrate the consequences of special relativity and made into a documentary called *The Ultimate Speed*. Electrons were accelerated in a linear accelerator to as high a speed as possible, and the speed of the electrons was measured as a function of their kinetic energy. The concepts of energy conservation and the work-energy theorem were implicit in the setup of the experiment: the electron's kinetic energy was determined by $\Delta K.E. = \int F dx$, where in this case F comes from an electrical force.

- 1.1. Using Newtonian mechanics, write down the kinetic energy K of the electron in terms of its speed v and its mass m_e . If we plotted this relationship on a graph where the x -axis was K and the y -axis was v^2 , this would be a line: what is its slope?

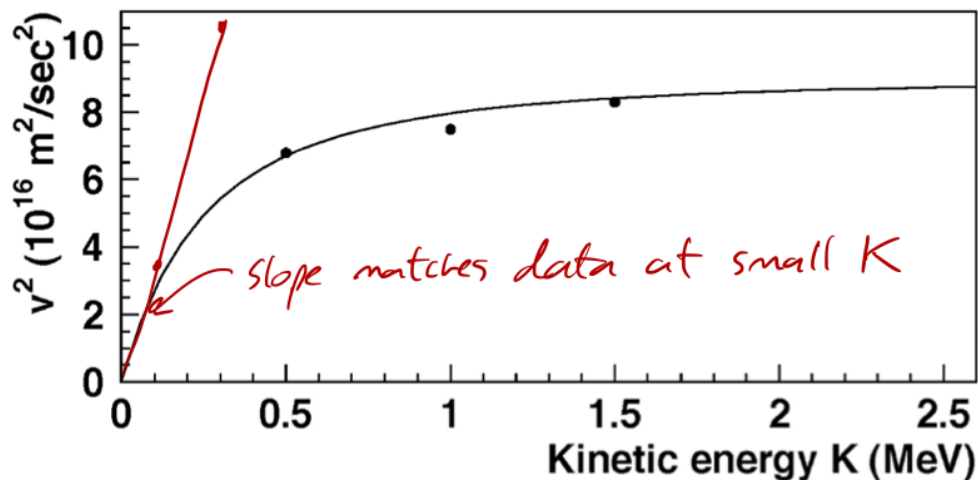
Let's write the kinetic energy with the following substitutions: $K \rightarrow x, v^2 \rightarrow y$.

$$K = \frac{1}{2}m_e v^2 \implies x = \frac{1}{2}m_e y \implies y = \frac{2}{m_e}x.$$

This expression is in slope-intercept form and we can read off that the slope is $\frac{2}{m_e}$.

- 1.2. The actual experimental data is shown in Fig. 1 (three data points are shown for illustration, but the curve is a fit to the full dataset). The units on the x -axis are MeV (mega electronvolts), which are related to more familiar SI units by $1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J}$. Draw your curve from part (a) on this plot, using the electron mass $m_e = 9.1 \times 10^{-31} \text{ kg}$. Where does this result agree with the data, and where is it in obvious disagreement?

Since we know the Newtonian result is a line, we only need two points: let's choose $K = 0.1 \text{ MeV}$ and $K = 0.3 \text{ MeV}$, which (using $v^2 = \frac{2}{m_e}K$ give $v^2 = 3.5 \times 10^{16} \text{ m}^2/\text{s}^2$ and $v^2 = 10.5 \times 10^{16} \text{ m}^2/\text{s}^2$, respectively. Sketching this on the plot (see below), we see that the Newtonian result matches the data at very small K , but quickly diverges from the data by $K = 0.2 \text{ MeV}$.



- 1.3. The fact that the experimental data levels off means that it is harder and harder to increase the speed of the electrons even after applying more and more kinetic energy, and they seem to approach a limiting speed. What is the numerical value of this speed? Why does this make sense given what you already know about special relativity?

The data seems to approach the maximum squared speed of about $9 \times 10^{16} \text{ m}^2/\text{s}^2$. The squared speed of light is about $c^2 \approx 9 \times 10^{16} \text{ m}^2/\text{s}^2$. This makes sense because an electron has a nonzero mass, and hence, it can never move at c . For a massive particle, its kinetic energy is given by $K = (\gamma - 1)mc^2$ where $\gamma = \frac{1}{\sqrt{1-(v/c)^2}}$ which diverges to infinity when $v = c$, and hence, c^2 is a strong upper bound for the squared speed.

Part 2: Explaining the data with special relativity

Recall that we derived the formula $K = (\gamma - 1)mc^2$ in lecture by applying the work-energy theorem. You will now show that this formula can explain the experimental data from Part 1, *and* that it reduces to the Newtonian result in the appropriate limit. You will need the Taylor series

$$(1 + x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha - 1)}{2}x^2 + \mathcal{O}(x^3).$$

- 2.1. Rearrange the relativistic kinetic energy formula to find v^2 as a function of K . Plug in a few values for K between 0 and 2.5 MeV and draw the corresponding values of v^2 on the experimental data plot (reproduced again below): how well do they agree?

Let's find $v^2(K)$.

$$K = (\gamma - 1)mc^2 \implies \frac{K}{mc^2} + 1 = \gamma.$$

Substitute in the definition of gamma and squaring gives

$$\left(\frac{K}{mc^2} + 1 \right)^2 = \frac{1}{1 - (v/c)^2}$$

Taking the reciprocal of both sides and solving for v gives

$$v^2(K) = \frac{K(K + 2mc^2)}{(K + mc^2)^2} c^2.$$

We once again can stick to SI units by plugging in the electron's mass in units of kg, the speed of light in units of m/s, and the kinetic energy in units of J. The table of values is shown below. These match the data extremely well.

K (MeV)	v^2 ($10^{16} \text{ m}^2/\text{s}^2$)
1	7.97
1.5	8.42
2	8.62
2.5	8.74

- 2.2. To see the limiting speed, take the limit of your formula as $K \rightarrow \infty$. Do you recover your answer from problem 1.3?

If we directly plug in $K \rightarrow \infty$, we arrive at an indeterminate form:

$$\lim_{K \rightarrow \infty} \frac{K(K + 2mc^2)}{(K + mc^2)^2} c^2 \rightarrow \frac{\infty(\infty + c^2)}{(\infty + c^2)^2} c^2 \rightarrow c^2 \frac{\infty}{\infty}.$$

We can use L'Hopital's Rule to differentiate the numerator and denominator with respect to K to see that

$$\lim_{K \rightarrow \infty} \frac{K(K + 2mc^2)}{(K + mc^2)^2} c^2 \approx \lim_{K \rightarrow \infty} \frac{2(K + c^2m)}{2(K + c^2m)} c^2 = \frac{2c^2}{2} = c^2.$$

Hence, we recover the same limit as before. (An alternate solution, which is less formal but is often used in physics, is to note that the highest power of K appearing in the numerator and denominator will dominate the limit as $K \rightarrow \infty$. Both numerator and denominator have K^2 with a coefficient of 1, so the limit is $(K^2/K^2)c^2 = c^2$.)

- 2.3. To recover Newtonian mechanics, take the opposite limit: Taylor-expand $(\gamma - 1)mc^2$ in the small dimensionless parameter v^2/c^2 , and check that you recover your result from problem 1.1 to leading order in v^2/c^2 .

We would like to put $(\gamma - 1)mc^2$ in the form of $(1 + x)^\alpha$ for some α and $x = -(v/c)^2$.

$$(\gamma - 1)mc^2 = \gamma mc^2 - mc^2 = (1 + x)^{-1/2} mc^2 - mc^2.$$

Now we can Taylor-expand $(1 + x)^{-1/2}$ and use $\alpha = -\frac{1}{2}$:

$$(1 + x)^{-1/2} \approx 1 + \alpha x + \frac{\alpha(\alpha - 1)x^2}{2} = 1 - \frac{1}{2}x + \frac{3}{8}x^2.$$

Now, we consider the kinetic energy and find it to leading order in $(v/c)^2$:

$$\begin{aligned} K &= mc^2 \left(1 + \frac{1}{2} \left(\frac{v}{c} \right)^2 + \frac{3}{8} \left(\frac{v}{c} \right)^4 \right) - mc^2 \\ \implies K &\approx \frac{1}{2} \left(\frac{v}{c} \right)^2 mc^2 = \frac{1}{2} mv^2 + \mathcal{O}(v^4/c^4) \end{aligned}$$

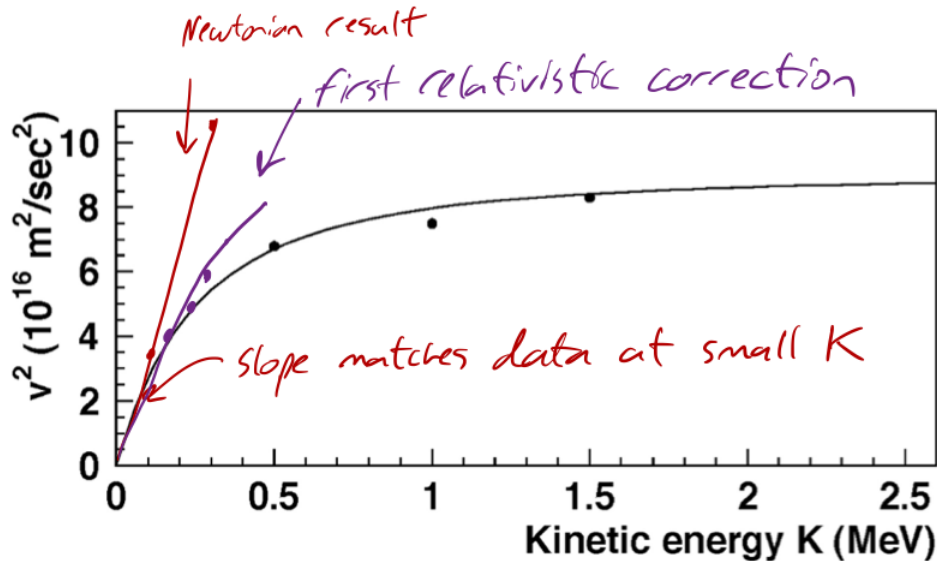
Thus, we recover the correct classical result when we consider the relativistic kinetic energy to leading order in $(v/c)^2$.

- 2.4. As we take more and more terms in the Taylor series, we can approach the full special relativistic result. Show this by plotting the first relativistic correction to the Newtonian formula on the plot from problem 2.1. Your answer should take the form $K = \frac{1}{2}m_e v^2 + C v^4$, so your job is to find the constant C . At approximately what speed v does the result begin to differ significantly from the data?

Taking the next term in the Taylor series from 2.3 above, we want to plot the first relativistic correction to the kinetic energy,

$$K = \frac{1}{2}m_e v^2 + \frac{3}{8}m_e \frac{v^4}{c^2}.$$

Matching this to the form $K = \frac{1}{2}m_e v^2 + C v^4$, we conclude that $C = \frac{3m_e}{8c^2}$. In this form it's easiest to pick values of v^2 and find the corresponding values of K . Taking $v^2 = 4, 5, 6, 7 \times 10^{16} \text{ m}^2/\text{s}^2$ Drawing this on the plot yields the purple curve below. We see that this approximation matches the data very well up to a speed of about $5 \times 10^{16} \text{ m}^2/\text{s}^2$.



- 2.5. Since K is small whenever v is small, we could just as well have performed a Taylor expansion in K . However, K is not dimensionless, so we have to compare it to something else with units of energy. A good choice is the electron rest energy $E_0 = m_e c^2$. Show that a Taylor expansion of your formula from problem 2.1 in the small parameter K/E_0 gives the correct Newtonian result $v^2 = 2K/m_e$. The implication of this calculation is that Newtonian mechanics is a reasonable approximation whenever the kinetic energy of a particle is small compared to its rest energy.

Let $x = \frac{K}{mc^2}$. Factoring out some factors of $E_0 = mc^2$, our formula from 2.1 can be re-written as:

$$v^2(x) = \frac{x E_0 (x E_0 + 2 E_0)}{(x E_0 + E_0)^2} c^2 = \frac{x(x+2)}{(1+x)^2} c^2.$$

Rather than use the definition of the Taylor series, let's just use our expansion on the denom-

inator, and then multiply by the numerator and collect powers of x :

$$\begin{aligned} (1+x)^{-2} &\approx 1-2x \\ \implies \frac{x(x+2)}{(1+x)^2}c^2 &\approx x(x+2)(1-2x)c^2 = 2xc^2 + \mathcal{O}(x^2). \end{aligned}$$

Plugging in for x , we have $v^2 \approx 2\frac{K}{mc^2}c^2 = \frac{2K}{m}$.

Part 3: A first look at relativistic kinematics with photons

We have already seen in several ways that it takes an infinite amount of energy to accelerate a massive particle to the speed of light. A massless particle would make no sense in Newtonian mechanics (since all of the kinematic formulas like $F = ma$ and $K = \frac{1}{2}mv^2$ would just give zero), but such particles are important features of special relativity. The most common example is the *photon*, which is a particle of light.¹

- 3.1. Take the ratio of the relativistic momentum and energy formulas to derive a useful formula for the velocity of a particle: $p/E = v/c^2$. Notice that this formula does not involve the mass!

$$\frac{p}{E} = \frac{\gamma mv}{\gamma mc^2} = \frac{v}{c^2}$$

- 3.2. Suppose we boldly assert that $p/E = v/c^2$ still applies when $v = c$. In that case, we could still assign an energy and a momentum to a photon, related by $p_{\text{photon}} = E_{\text{photon}}/c$. Rearrange this formula to show $E_{\text{photon}}^2 - p_{\text{photon}}^2 c^2 = 0$.

Letting $v = c$, we see that

$$\begin{aligned} \frac{p}{E} &= \frac{v}{c^2} = \frac{1}{c} \\ \implies pc &= E \\ \implies p^2 &= \frac{E^2}{c^2} \\ \implies E^2 - p^2 c^2 &= 0 \end{aligned}$$

- 3.3. Use the relativistic momentum and energy formulas for a massive particle to show that

$$E^2 - p^2 c^2 = m^2 c^4.$$

This is probably the single most important formula for the remainder of the course! And by your work in problem 3.2, it also applies for a massless particle with $m = 0$.

¹If you're used to thinking of light as a wave, just wait for PHYS 486/487! Quantum mechanics tells us that all particles can behave like waves, and vice versa.

We want to use $p = \gamma mv$ and $E = \gamma mc^2$ to find $E^2 - p^2 c^2 = m^2 c^4$. With these substitutions, we see that

$$E^2 - p^2 c^2 = \gamma^2 m^2 c^4 - \gamma^2 m^2 v^2 c^2.$$

Note that $\gamma^2 = \frac{1}{1 - (\frac{v}{c})^2}$, so

$$E^2 - p^2 c^2 = \frac{m^2 c^2 (c^2 - v^2)}{1 - (\frac{v}{c})^2} = \frac{m^2 c^4 (c^2 - v^2)}{c^2 - v^2} = m^2 c^4.$$

Therefore, we recover that $E^2 - p^2 c^2 = m^2 c^4$.

- 3.4. Since a particle at rest has $E = E_0$, it can convert its rest energy into other forms of energy. In particular, there is an unstable subatomic particle called the π^0 (pronounced “pi-zero” or “neutral pion”) which decays into two photons. Use the results of problems 3.2 and 3.3 to determine the energies of the two photons in terms of the mass of the pion m_π . To do this, you can work in the rest frame of the pion, and assume that the photons’ momentum vectors are aligned with their velocity vectors (this will be fully justified in a few weeks). First apply momentum conservation, followed by energy conservation.

The initial energy is $E_i = m_\pi c^2$ which is just the rest energy of the pion. The two photons carry off equal amounts of the momentum in opposite directions due to momentum conservation. But since $E = pc$ for a photon, equal momentum means equal energy. Let the energy of one of the photons be E_γ ; then, the final energy is $E_f = 2E_\gamma$. Using energy conservation to set the initial and final energies equal, we find

$$E_\gamma = \frac{m_\pi c^2}{2}.$$

The rest energy of the pion is converted into two photons of equal energy as a result of the decay.